

INSTRUCTIONS TO THE CANDIDATES

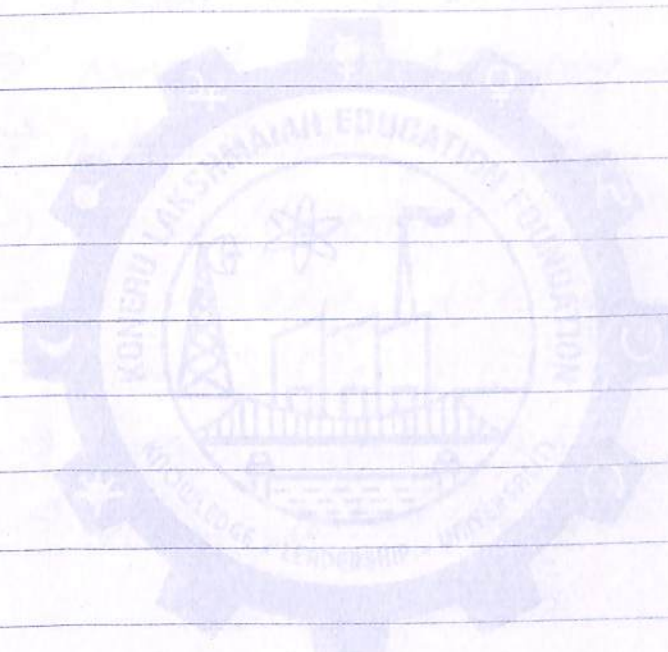
- ❖ Do not use sketch pens/ high –lighter pens/ color pencils etc., in underlining / highlighting certain points in the answer or while drawing sketches etc.,
- ❖ Check the Answer booklet thoroughly. The defective answer booklet may be returned to the invigilator before starting the examination.
- ❖ Ensure that the OMR Barcode sheet is properly secured to the answer booklet.
- ❖ Do not write on or tamper the Barcode. If you do so, your booklet will not be evaluated.
- ❖ The candidate is prohibited from writing his/ her NAME / ADDRESS/ PHONE NUMBER / EMAIL ID / Religion or any other matter on any part of the answer booklet / drawing sheet/ graph sheet/ addressing the examiner in any manner whatsoever in the answer booklet. The candidate is not supposed to tear a page in full or part of the booklet.

Non-adherence with this instruction shall result in candidate being booked under malpractice.

- ❖ Answer should be written on both sides of the paper. Candidate should write in all 25 lines in each page. It is not necessary to begin each answer in a fresh page.
- ❖ Candidate should write only question number in the margin.
- ❖ Candidate should return the answer booklet to the invigilator before leaving the examination hall.
- ❖ **ADDITIONAL SUPPLEMENT OF ANSWER SHEETS WILL NOT BE ISSUED.**

INSTRUCTIONS TO THE EXAMINERS

- ❖ Invigilators should ensure that the above instructions are strictly followed by the candidates.
- ❖ After verifying the particulars on the OMR sheet, invigilator should sign in the box specified.
- ❖ Do not crumple or cut the sides of the sheets.
- ❖ Do not write, mark or damage the timing tracks or Barcode.



1 A

UNIX commands.

- LS → list everything in the directory
- PING → Ping an ip address sending packets
- PWD → Present working directory
- MAN → Manual for the command.
- MKDIR → make a directory
- RM → remove/delete files
- RMDIR → remove empty directory.
- TOUCH → creating files
- CAT → Displaying contents of a file.
- NANO → change contents of a file.

1 B

Types of OS

- | | |
|----------------------|------------------|
| • Banking OS | • Open source OS |
| • Mobile OS | • Proprietary OS |
| • Accounting OS | |
| • Crypto OS | |
| • Multiprocessing OS | |

1 C

Features of UNIX.

- Compact
- Portable
- Accountable
- Open source
- Multi processing

1 D

Free Space Management.

- Free space management is that of the utilizing of free space for various things like multi processing.

Or even freeing space from non-important tasks and using it wisely for other programs.

1 E

Segmentation. Array data.

→ The grouping of instructions into segments.

→ It helps the process to run faster.

1 F

#include <stdlib.h>

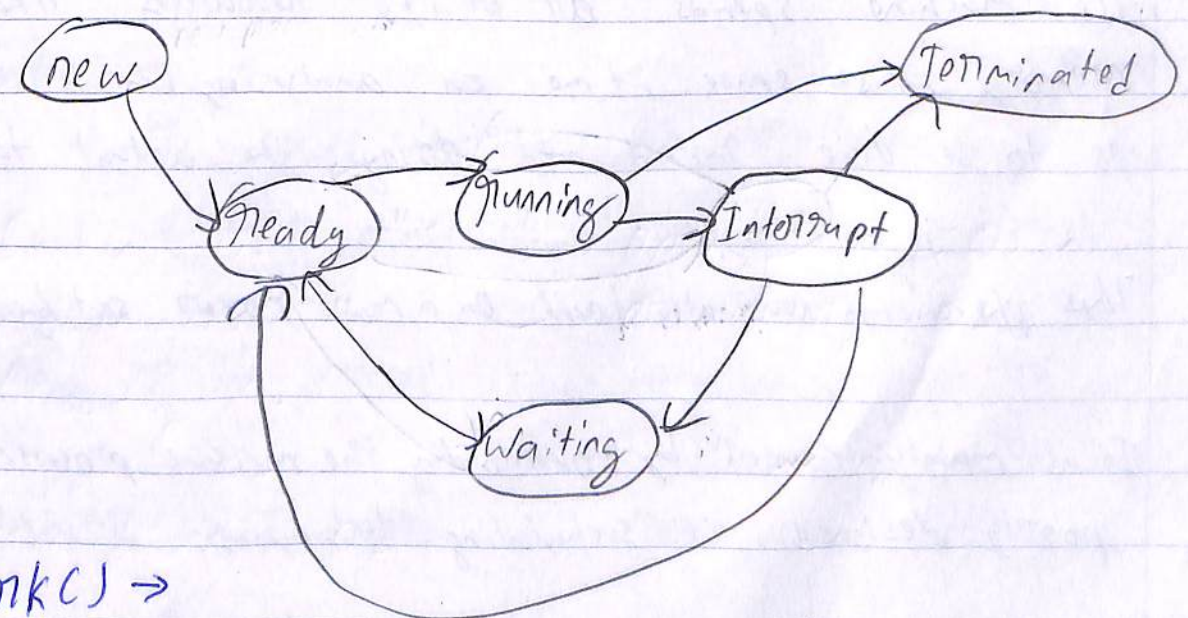
The role of free() is to free the memory allocated for a specific address.

With this dynamic memory allocated freed the memory can be used again.

2 A

Process

→ A process is a running program allocated in the memory.



fork() →

It creates a copy of the process.

2 B

Multi-Level Feedback.

It is the process of getting feedback from the task completed on various depths or levels. With given said feedbacks, we can improve the things accomplished by the processes.

2 D

Thrashing.

It is a state in which a machine is said to be 'procastinating'. The machine spends all of its resources like memory, disk space, time on analysing which task is to be done instead of doing the actual task.

This phenomenon was also found on a Mars rover sent by NASA.

It is mainly caused by overloading the machine processes and poorly designed CPU scheduling techniques.

4 A

Process.

→ A process is a running program in execution memory.

States

• New

→ A new process is formed

• Ready

→ The process is ready to be executed

• Waiting

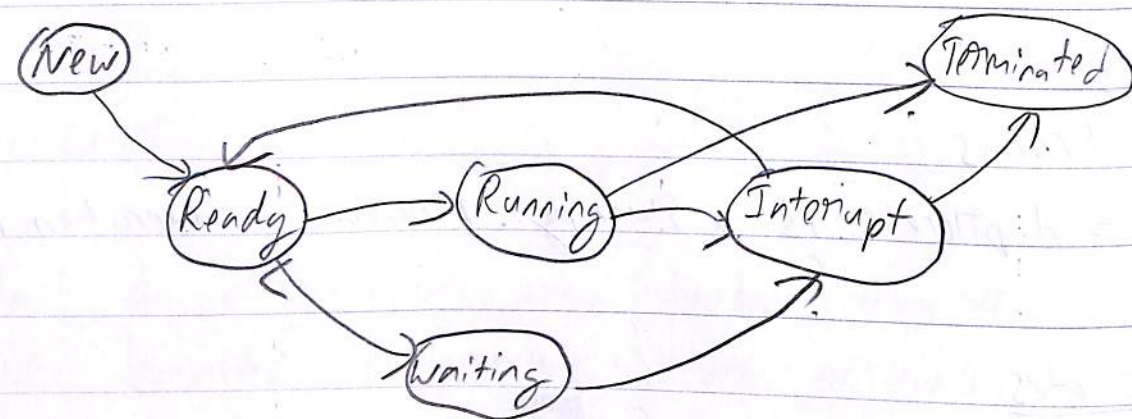
→ The process is to be executed waiting.

• Interrupt

→ The process is interrupted internally/externally and execution process is paused

• Running → The process is in execution.

• Terminated → The process is completed and killed.



4 B

FCFS, SJF

FCFS ~~First~~ First Come First Serve.

→ First come first serve is when the a process is executed based on the order of Arrival. F/FO Like a QUEUE

SJF Shortest Job First.

→ Shortest job first is when the CPU compares which job can be completed first and let that process be executed. This may also lead to starving

5 process.

FCFS

P	AT	BT	CT	CT-AT	TAT-BT
				TAT	WT
P ₁	0	7	07	7	0
P ₂	1	5	12	11	6
P ₃	2	3	15	13	10
P ₄	6	2	17	11	9
P ₅	12	7	24	12	5

$$AVG TAT = \frac{7+11+13+11+12}{5}$$

$$= \frac{54}{5} = 10.8$$

$$AVG WT = \frac{0+6+10+9+5}{5} = \frac{30}{5} = 6$$

P1	P2	P3	P4	P5
0	7	12	15	17
24				

5 process / SJF /
P. / AT / BT / CT / TAT / WT

5 process STF

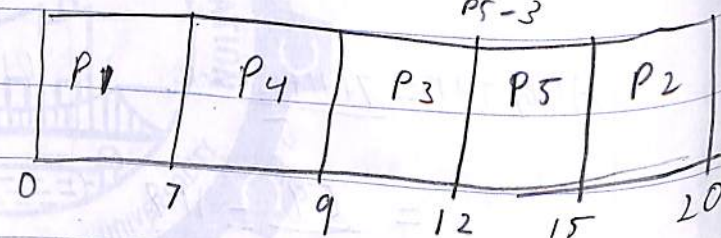
P	AT	BT	CT	CT-AT	TAT-BT	WT
P ₁	0	7	7	7	0	0
P ₂	1	5	9	8	3	3
P ₃	2	3	12	10	7	7
P ₄	6	2	15	9	7	7
P ₅	12	3	20	8	5	5

T=0
P₁ - 7

T=7
P₂ - 5
P₃ - 3
P₄ - 2

T=9
P₂ - 5
P₃ - 3

T=12
P₂ - 5
P₅ - 3



$$\text{AVG TAT} = \frac{7+8+10+9+8}{5} = 36$$

$$= \frac{42}{5} = 8.4$$

$$\text{AVG WT} = \frac{0+3+7+7+5}{5} = \frac{22}{5} = 4.4$$

FCFS	FCFS	STF
AVG TAT	11.8	8.4

AVG WT	6	4.4
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5 B

Least Recently Used Algorithm.

Frames = 4

7 0 1 2 0 3 0 4 2 3 0 3 2

7	0	1	2	hit	3	hit	4	hit	hit	3	hit	hit
7	7	7	7	7	3	3	3	3	3	3	3	3
	0	0	0	0	0	0	4	4	4	4	4	4
		1	1	1	1	1	1	1	1	0	0	0
			2	2	2	2	2	2	2	2	2	2
				hit		hit		hit	hit		hit	hit

5 A

Memory.

It is within memory that everything is stored.

All of the processes and execution is done here

Without memory there wouldn't be such thing as OS.

I don't know please help

$P = 0.05$

